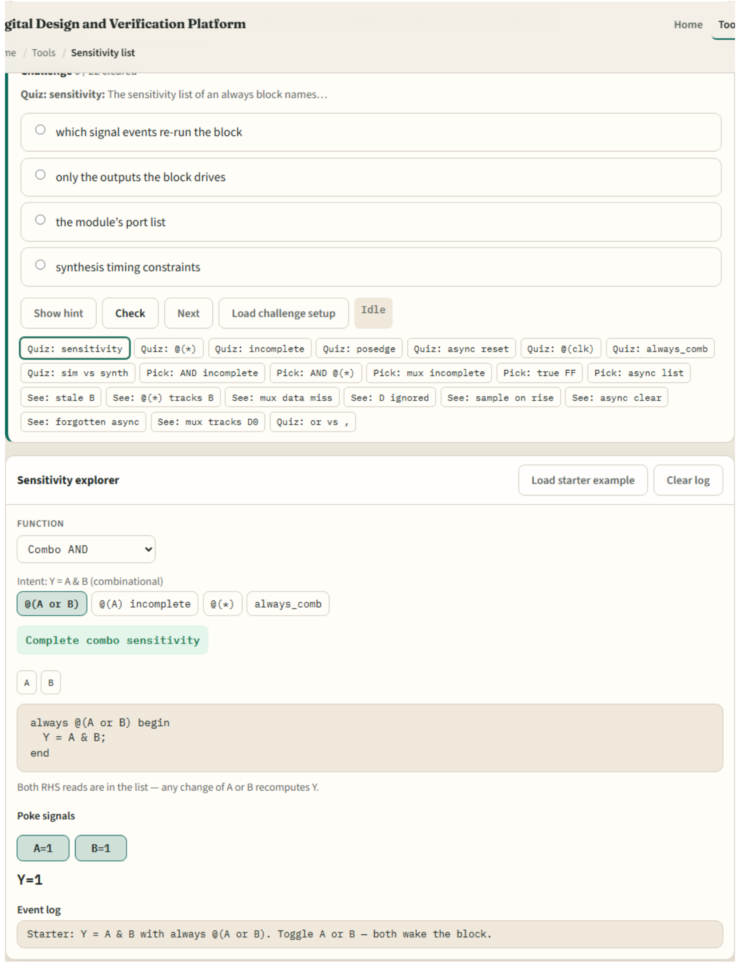
Sensitivity lists

An always block does not run on every tick of simulation time

What wakes an always block

- For combinational logic
- Classic Verilog wrote at open-paren A or B close-paren
- If you list only A but also read B
- For sequential logic, posedge clk means the block runs only on a rising clock edge

Browser lab



Sensitivity list lab starter

Real Verilog practice

```
wsl — module06-sensitivity-list/examples/navigation

# Track A — real Unix

# real Verilog session (Track A)

$ cat examples/sensitivity/sensitivity.v
// sensitivity.v — combo list vs clock edge
module sens_demo(
  input      A,
  input      B,
  input      clk,
  input      D,
  output reg Y,
  output reg Q
);
  always @(A or B)
    Y = A & B;

  always @(posedge clk)
    Q <= D;
endmodule

$ iverilog -t null examples/sensitivity/sensitivity.v
(syntax check passed)
```

Real Verilog — combo and clocked sensitivity

Pitfalls to watch

- Do not hand-write or lists and forget a signal you read
- Do not use level-sensitive at open-paren clk close-paren when you mean a flip-flop
- At-star fixes combo lists in modern Verilog
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter, try the incomplete list, and poke the flip-flop scenario
- On real Verilog
- When you are ready, take the short quiz, then continue to latch risk in the next module